

POWERFACTORY

PowerFactory 2024

Technical Reference

Series Capacitor

ElmScap

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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March 28, 2024
PowerFactory 2024
Revision 2

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1 General Description

PowerFactory provides both an AC and a DC series capacitor model. The AC series capacitor model may be used for various applications which require the connection of the series capacitance to an AC terminal, such as:

- Series compensation of transmission lines
- Filter capacitance

The DC series capacitor model may be used for applications which require connection to a DC terminal, such as:

- Detailed voltage source converter modelling
- FACTS controller modelling

Figure 1.1 shows the basic scheme of the series capacitor.

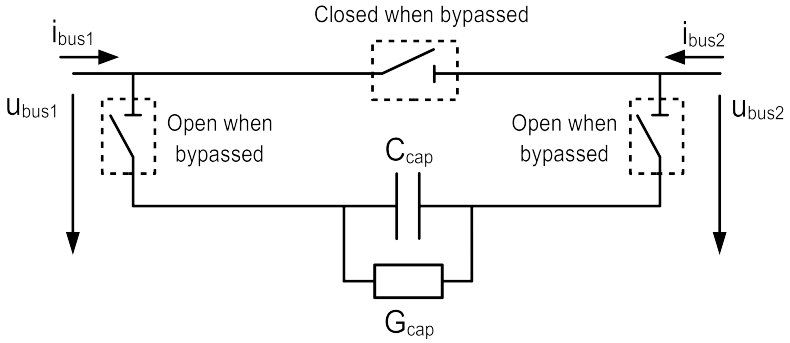


Figure 1.1: Basic Scheme

In case of the option *Bypassed* is enabled the corresponding bypass switch is closed and the series capacitor is isolated by opening the corresponding separators.

Losses are modelled as a conductance in parallel with the capacitor.

The model uses a nonlinear resistance in parallel with the capacitance for modelling protection of series capacitances by a MOV-bypass. Under normal operating conditions, it is assumed that the parallel resistance has no influence on the capacitance. It is therefore only considered in EMT simulations.

The rated voltage and rated current (or rated power) can be specified, as well as a *Thermal Ratings*. The *System Type* can be set to either *AC* or *DC* and the *Phases* to either *1* or *3*. The three-phase model is illustrated in Figure 1.2, and the single-phase model is illustrated in Figure 1.3.

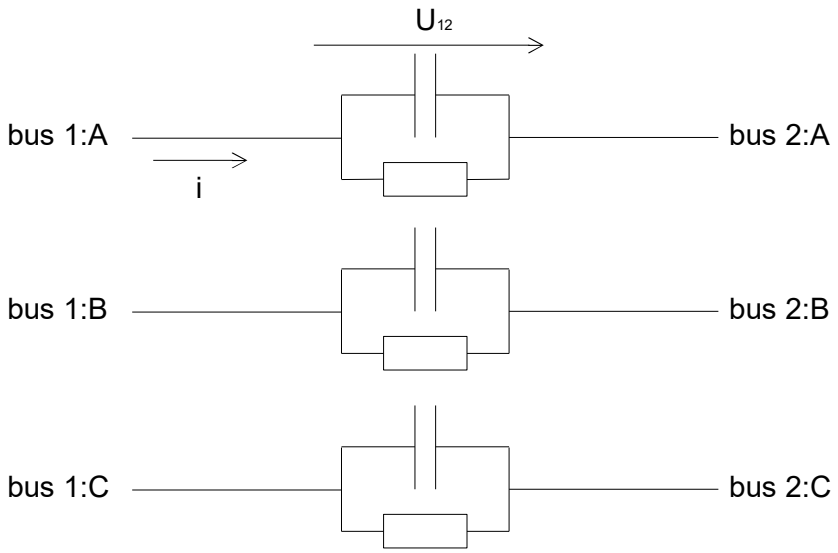


Figure 1.2: Three-phase model

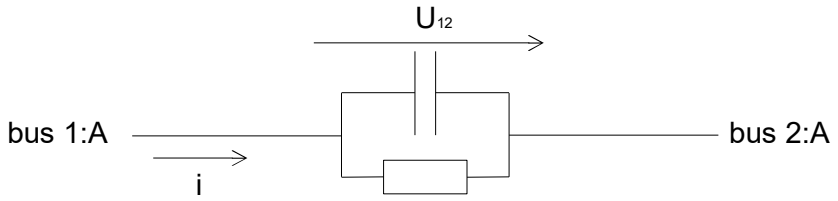


Figure 1.3: Single-phase model

1.1 Input Parameter

1.1.1 Admittance

The series capacitor is modelled with a capacitance in parallel with a conductance, representing the losses.

Capacitance The capacitance of the capacitor can be entered with different input options (only for system type: AC and AC/BI):

- Capacitance in F
- Suscepteance in S
- Reactance in Ω

When using the three-phase model in conjunction with the *Unbalanced* option on the *Load Flow* tab of the series capacitor element dialog, the user may specify a susceptance on a per-phase

basis. In this case, all balanced calculation methods are using the average capacitance of the the three phase as positive sequence capacitance $C_{cap} = 1/3(C_{a,cap} + C_{b,cap} + C_{c,cap})$.

For a DC series capacitor, only the Capacitance C in F can be entered. Other input options are not supported.

Parameter	Description	Unit	Symbol
For Balanced *option:*			
ccap	Capacitance, C	F	C_{cap}
bcap	Susceptance, B	S	B_{cap}
xcap	Reactance 1/B	Ω	X_{cap}
For Unbalanced *option:*			
ccap_a	Capacitance, C, phase A	F	$C_{a,cap}$
ccap_b	Capacitance, C, phase B	F	$C_{b,cap}$
ccap_c	Capacitance, C, phase C	F	$C_{c,cap}$
bcap_a	Susceptance, B, phase A	S	$B_{a,cap}$
bcap_b	Susceptance, B, phase B	S	$B_{b,cap}$
bcap_c	Susceptance, B, phase C	S	$B_{c,cap}$
xcap_a	Reactance 1/B, phase A	Ω	$X_{a,cap}$
xcap_b	Reactance 1/B, phase B	Ω	$X_{b,cap}$
xcap_c	Reactance 1/B, phase C	Ω	$X_{c,cap}$

Table 1.1: Suceptance, Capacitance parameter

Losses Losses in the series capacitor can be modelled as a conductance in parallel with the capacitor. The conductance is defined directly (parameter *gcap*) if the system type is DC and through the *Loss factor, tan(delta)* parameter (*tandc*) for system type AC and AC/BI. The relation between conductance used in the calculations, susceptance and loss factor is:

$$gcap = bcap \cdot tandc$$

for *Balanced* option, and:

$$gcap_a = bcap_a \cdot tandc$$
$$gcap_b = bcap_b \cdot tandc$$
$$gcap_c = bcap_c \cdot tandc$$

for *Unbalanced* option.

1.1.2 Zero Sequence Admittance

A zero sequence admittance can be defined through the parameters *G0/G1 ratio* and *B0/B1 ratio*. The zero sequence admittance is considered only in unbalanced calculations. The definition of the zero sequence admittance is supported only for the *Balanced* input mode option and for system type AC. The zero sequence conductance and susceptance are given as:

$$G_{0,cap} = G0toG1 \cdot G_{cap}$$
$$B_{0,cap} = B0toB1 \cdot B_{cap}$$

1.1.3 Input mode of the rating parameter

For the definition of the rating parameter there are two different input modes available:

a) Rated Voltage in kV and Rated Power in MVA

b) Rated Voltage in kV and Rated Current in kA

For a three phase capacitor the rated current is calculated as:

$$I_r = \frac{S_r}{\sqrt{3} \cdot U_r} \quad \text{kA}$$

for an AC single phase capacitor:

$$I_r = \frac{\sqrt{3} \cdot S_r}{U_r} \quad \text{kA}$$

for an AC/BI single phase capacitor:

$$I_r = \frac{2 \cdot S_r}{U_r} \quad \text{kA}$$

and for a DC single phase capacitor:

$$I_r = \frac{S_r}{U_r} \quad \text{kA}$$

where:

Parameter	Description	Unit	Symbol
ucn	Rated Voltage (Line-Line)	kV	U_r
Sn	Rated Power	MVA	S_r
Curn	Rated Current	kA	I_r
systp	System Type (AC, DC, AC/BI)		

Table 1.2: Rating parameter

The rated current I_r is the base for p.u. current quantities.

1.1.4 Susceptance B_{cap} <=> Capacitance C_{cap} <=> Reactance relation

$$B_{cap} = 2 \cdot \pi \cdot f_{nom} \cdot C_{cap}$$
$$X_{cap} = 1/B_{cap}$$

where:

- C_{cap} : Capacitance in F
- B_{cap} : Susceptance in S
- X_{cap} : Reactance in Ω
- f_{nom} : Nominal frequency of the grid in Hz

2 Load Flow Analysis

2.1 AC Model

In all steady-state analysis functions the capacitance is modelled by its nominal frequency behaviour.

In case of a balanced load flow:

$$\underline{I}_{bus1} = (G_{cap} + jB_{cap}) \cdot \underline{U}_{12} \quad (1)$$

$$\underline{I}_{bus1} + \underline{I}_{bus2} = 0 \quad (2)$$

In case of an unbalanced load flow and with the *Balanced* input option:

$$\begin{aligned} \underline{I}_{1,bus1} &= (G_{cap} + jB_{cap}) \cdot \underline{U}_{1,12} \\ \underline{I}_{1,bus1} + \underline{I}_{1,bus2} &= 0 \\ \underline{I}_{2,bus1} &= (G_{cap} + jB_{cap}) \cdot \underline{U}_{2,12} \\ \underline{I}_{2,bus1} + \underline{I}_{2,bus2} &= 0 \\ \underline{I}_{0,bus1} &= (G_{0,cap} + jB_{0,cap}) \cdot \underline{U}_{0,12} \\ \underline{I}_{0,bus1} + \underline{I}_{0,bus2} &= 0 \end{aligned}$$

where $\underline{I}_{1,bus1}$, $\underline{I}_{2,bus1}$ and $\underline{I}_{0,bus1}$ are the positive, negative and zero sequence current at bus 1; $\underline{I}_{1,bus2}$, $\underline{I}_{2,bus2}$ and $\underline{I}_{0,bus2}$ are the positive, negative and zero sequence current at bus 2; $\underline{U}_{1,12}$, $\underline{U}_{2,12}$ and $\underline{U}_{0,12}$ are the positive, negative and zero sequence voltage difference between bus 1 and bus 2.

In case of an unbalanced load flow and with the *Unbalanced* input option:

$$\begin{aligned} \underline{I}_{a,bus1} &= (G_{a,cap} + jB_{a,cap}) \cdot \underline{U}_{a,12} \\ \underline{I}_{b,bus1} &= (G_{b,cap} + jB_{b,cap}) \cdot \underline{U}_{b,12} \\ \underline{I}_{c,bus1} &= (G_{c,cap} + jB_{c,cap}) \cdot \underline{U}_{c,12} \\ \underline{I}_{a,bus1} + \underline{I}_{a,bus2} &= 0 \\ \underline{I}_{b,bus1} + \underline{I}_{b,bus2} &= 0 \\ \underline{I}_{c,bus1} + \underline{I}_{c,bus2} &= 0 \end{aligned}$$

2.1.1 Calculation Quantities

Loading

The loading of the capacitor is calculated as follows:

$$loading = max\left(\frac{|I_{bus1}|}{I_{nom(bus1)}}, \frac{|I_{bus2}|}{I_{nom(bus2)}}\right) \cdot 100 \quad \%$$

where:

- $I_{nom(bus1)}$ is nominal current of the capacitor for terminal 1 in kA
- $I_{nom(bus2)}$ is nominal current of the capacitor for terminal 2 in kA
- I_{bus1} is magnitude of the current at terminal 1
- I_{bus2} is magnitude of the current at terminal 2

If no thermal ratings object is defined, the nominal currents are equal to the rated current of the capacitor $I_{nom(bus1)} = I_{nom(bus2)} = I_r$ (see 1.1.3).

If a thermal ratings object is used, the nominal currents are calculated using the parameter $ContRating$ and $U_{n(bus1)}$, $U_{n(bus2)}$ (nominal voltage in kV of the connected terminal 1 and 2) as:

- if the continuous rating is entered in MVA:
  - 3-phase capacitor:

$$I_{nom(bus1)} = ContRating / \left(\sqrt{3} \cdot U_{n(bus1)}\right) \quad \text{kA}$$
$$I_{nom(bus2)} = ContRating / \left(\sqrt{3} \cdot U_{n(bus2)}\right) \quad \text{kA}$$

  - AC single phase capacitor:

$$I_{nom(bus1)} = ContRating \cdot \sqrt{3} / U_{n(bus1)} \quad \text{kA}$$
$$I_{nom(bus2)} = ContRating \cdot \sqrt{3} / U_{n(bus2)} \quad \text{kA}$$

  - AC/BI single phase capacitor:

$$I_{nom(bus1)} = ContRating \cdot 2 / U_{n(bus1)} \quad \text{kA}$$
$$I_{nom(bus2)} = ContRating \cdot 2 / U_{n(bus2)} \quad \text{kA}$$

- if the continuous rating is entered in kA:

$$I_{nom(bus1)} = I_{nom(bus2)} = ContRating \quad \text{kA}$$

- if the continuous rating is entered in %:

$$I_{nom(bus1)} = I_{nom(bus2)} = ContRating / 100 \cdot I_r \quad \text{kA}$$

The nominal currents are generally the same ($I_{nom(bus1)} = I_{nom(bus2)}$), but not if the reactor is connected between different voltage levels ($U_{n(bus1)} \neq U_{n(bus2)}$).

Further details regarding thermal ratings object (*IntThrating*), see Technical Reference of the Thermal Ratings Definition

For an unbalanced load flow calculation the highest current of all phases is used.

Losses

The losses are calculated as follows:

Quantity	Unit	Description	Value
$Ploss$	MW	Losses (total)	$= P_{bus1} + P_{bus2}$
$Qloss$	Mvar	Reactive-Losses (total)	$= Q_{bus1} + Q_{bus2}$
$Plossld$	MW	Losses (load)	$= Ploss - Plossnld$
$Qlossld$	Mvar	Reactive-Losses (load)	$= Qloss - Qlossnld$
$Plossnld$	MW	Losses (no load)	$= 0$
$Qlossnld$	Mvar	Reactive-Losses (no load)	$= 0$

Table 2.1: Losses Quantities, AC-model

Voltage Drop

Quantity	Unit	Description	Value
du	p.u.	Voltage Drop	$= \lvert\underline{u}_{bus1}\rvert - \lvert\underline{u}_{bus2}\rvert$
$dupc$	%	Voltage Drop	$= du \cdot 100$
$dphiu$	deg	Voltage Drop Angle	$= \phi_{u,bus1} - \phi_{u,bus2}$
$du1$	p.u.	Positive Sequence Voltage Drop	$= \lvert\underline{u1}_{bus1}\rvert - \lvert\underline{u1}_{bus2}\rvert$
$du1pc$	%	Positive Sequence Voltage Drop	$= du1 \cdot 100$
$dphiu1$	deg	Positive Sequence Voltage Drop Angle	$= \phi_{u1,bus1} - \phi_{u1,bus2}$

Table 2.2: Voltage Drop Quantities, AC-model

where u_{bus1} and u_{bus2} are the amplitudes of the corresponding terminal voltage in p.u. based on the rated voltage of the terminal. $\phi_{u,bus1}$ and $\phi_{u,bus2}$ the terminal voltage angle in deg. For an unbalanced load flow du, $dupc$, and $dphiu$ are available per phase (e.g. $c : dupc : B$).

2.2 AC model for linear DC Load Flow

Only the 3-phase series capacitor is considered for the DC Load Flow:

$$(\phi_{bus1} - \phi_{bus2})/X_{cap} = P_{bus1}$$
$$P_{bus1} + P_{bus2} = 0$$

where $X_{cap} = -1/B_{cap}$ and ϕ_{bus1} is the voltage angle on terminal i, ϕ_{bus1} angle on terminal j.

2.2.1 Calculation Quantities

Loading

The loading of the reactor is calculated as follows:

$$loading = max\left(\frac{|P_{bus1}|}{P_{nom(bus1)}},\ \frac{|P_{bus1}|}{P_{nom(bus2)}}\right) \cdot 100\ \%$$

where:

- P_{bus1} : Active power at terminal 1
- P_{bus2} : Active power at terminal 2
- $P_{nom(bus1)}$: Nominal power at terminal 1
- $P_{nom(bus2)}$: Nominal power at terminal 2

The nominal power is determined as follow when no thermal ratings object is defined:

- $P_{nom(bus1)} = \sqrt{3} \cdot U_{n(bus1)} \cdot I_r$
- $P_{nom(bus2)} = \sqrt{3} \cdot U_{n(bus2)} \cdot I_r$

with the rated current of the capacitor I_r (see 1.1.3).
$U_{n(bus1)}$, $U_{n(bus2)}$ is the nominal voltage in kV of the connected terminals.

If a thermal ratings object is used, the nominal power is calculated using the parameter $ContRating$ and $U_{n(bus1)}$ and $U_{n(bus2)}$ (nominal voltage in kV of the connected terminal 1 and 2) as:

- if the continuous rating is entered in MVA:

$$P_{nom(bus1)} = P_{nom(bus2)} = ContRating \quad \text{MW}$$

- if the continuous rating is entered in kA:

$$P_{nom(bus1)} = \sqrt{3} \cdot U_{n(bus1)} \cdot ContRating \quad \text{MW}$$
$$P_{nom(bus2)} = \sqrt{3} \cdot U_{n(bus2)} \cdot ContRating \quad \text{MW}$$

- if the continuous rating is entered in %:

$$P_{nom(bus1)} = \sqrt{3} \cdot U_{n(bus1)} \cdot ContRating/100 \cdot I_r \quad \text{MW}$$
$$P_{nom(bus2)} = \sqrt{3} \cdot U_{n(bus2)} \cdot ContRating/100 \cdot I_r \quad \text{MW}$$

The nominal powers are generally the same ($P_{nom(bus1)} = P_{nom(bus2)}$), but not if the reactor is connected between different voltage levels ($U_{n(bus1)} \neq U_{n(bus2)}$).

Further details regarding thermal ratings object (*IntThrating*), see Technical Reference of the Thermal Ratings Definition

Losses

Losses are not calculated in the linear DC Load Flow.

2.3 DC Model

Under DC-conditions, a capacitance represents an open-circuit, hence:

$$I_{bus1} = G_{cap} \cdot U_{12} \tag{3}$$

2.3.1 Calculation Quantities

Loading

The loading of the capacitor is calculated as follows:

$$loading = max\left(\frac{|I_{bus1}|}{I_{nom(bus1)}}, \frac{|I_{bus2}|}{I_{nom(bus2)}}\right) \cdot 100\ \%$$

where:

- $I_{nom(bus1)}$ is nominal current of the capacitor for terminal 1 in kA
- $I_{nom(bus2)}$ is nominal current of the capacitor for terminal 2 in kA
- I_{bus1} is magnitude of the current at terminal 1
- I_{bus2} is magnitude of the current at terminal 2

If no thermal ratings object is defined, the nominal currents are equal to the rated current of the capacitor $I_{nom(bus1)} = I_{nom(bus2)} = I_r$ (see 1.1.3).

If a thermal ratings object is used, the nominal currents are calculated using the parameter $ContRating$ and $U_{n(bus1)}$, $U_{n(bus2)}$ (nominal voltage in kV of the connected terminal 1 and 2) as:

- if the continuous rating is entered in MVA:

$$I_{nom(bus1)} = ContRating/U_{n(bus1)} \quad \text{kA}$$
$$I_{nom(bus2)} = ContRating/U_{n(bus2)} \quad \text{kA}$$

- if the continuous rating is entered in kA:

$$I_{nom(bus1)} = I_{nom(bus2)} = ContRating \quad \text{kA}$$

- if the continuous rating is entered in %:

$$I_{nom(bus1)} = I_{nom(bus2)} = ContRating/100 \cdot I_r \quad \text{kA}$$

The nominal currents are generally the same ($I_{nom(bus1)} = I_{nom(bus2)}$), but not if the reactor is connected between different voltage levels ($U_{n(bus1)} \neq U_{n(bus2)}$).

Further details regarding thermal ratings object (*IntThrating*), see Technical Reference of the Thermal Ratings Definition

Losses

The losses are calculated according to Table 2.1.

3 Short-Circuit Calculation

3.1 AC Model

In general, series capacitors configured as AC capacitor models are considered with their nominal frequency behaviour as described in Section 2.1. Depending on the selected short-circuit method, additional considerations apply as described below.

Please refer to the *PowerFactory* User Manual for detailed information about the different configuration options of the short-circuit command.

3.1.1 VDE/IEC Short-Circuit

Note: Series capacitors configured as single-phase models are not considered in any *PowerFactory* VDE/IEC short-circuit calculation.

Depending on the calculation method used for peak or decaying short-circuit current, the following applies for the calculation of the corresponding R/X ratio:

Single-fed and multiple single-fed short-circuits Capacitors are considered when determining the R/X ratio of the feeding branch;

Multiple-fed short-circuits Capacitors are considered if the selected method is “B”, but neglected (i.e. bypassed) otherwise.

3.1.2 Complete Short-Circuit

Series capacitors are always neglected (i.e. bypassed) when calculating the R/X ratio for peak or decaying short-circuit current.

The option *Consider bypass circuit* allows for integrated protective circuits (such as a spark gap or metal oxide varistor) to be considered automatically during the short-circuit calculation. Voltage and current thresholds for activating the bypass can be enabled separately or together. The phase current is compared with the current threshold (*e:shcIprot*) and the voltage across the series capacitor with the voltage threshold (*e:shcVprot*).

If at least one threshold is enabled, the short-circuit calculation is carried out as follows:

1. Calculate the subtransient short-circuit without bypass;
2. Evaluate current and/or voltage against the thresholds and activate the bypass if necessary;
3. If the bypass is activated, the subtransient short-circuit is re-calculated.

The state of the bypass determined for the subtransient calculation is retained for the transient calculation.

For unbalanced short-circuit calculations, it is sufficient for one phase to exceed the threshold to trigger the bypass for the whole series capacitor.

The option *Consider Metal Oxide Varistor* enables a linearised MOV model according to [1]. This model only has an effect if the short-circuit calculation has the *Current Iteration* option enabled.

If the current through the capacitor exceeds the set value ($I \geq 0{,}98 \cdot I_{pr}$) the MOV is switched into conducting mode. The factor of $0{,}98$ is used to account for the fact that MOVs always conduct a small current and can not be considered as discrete switches. If the MOV is conducting, the following equations account for the voltage over the capacitor:

$$\underline{I} = \underline{Z}_{lin} \cdot (\underline{U}_i - \underline{U}_j) = (R^{'} - j \cdot X_C^{'}) \cdot (\underline{U}_i - \underline{U}_j) \quad (4)$$

$$R^{'} = X_C * (0{,}0745 + 0{,}49 \cdot e^{-0{,}243 \cdot i_{pr}} - 35 \cdot e^{-5 \cdot i_{pr}} - 0{,}6 \cdot e^{-1{,}4 \cdot i_{pr}}) \quad (5)$$

$$X_C^{'} = X_C * (0{,}1010 - 0{,}005749 \cdot i_{pr} + 2{,}088 * e^{-0{,}8566 \cdot i_{pr}}) \quad (6)$$

Where:

$\underline{I}$ is the complex current through the capacitor in kA

$\underline{Z}_{lin}$ is the linearised impedance in Ω (*c:Zlin*)

$\underline{U}_i$, $\underline{U}_j$ are the complex line-ground voltages at bus i and bus j respectively in kV

X_C is the nominal reactance of the capacitor in Ω (see Section 1.1.1)

i_{pr} is the current through the capacitor referred to the protective level current

$$i_{pr} = \frac{|\underline{I}|}{I_{pr}}$$

I_{pr} is the protective level current input value in kA (*e:shcMovIpr*)

The equations for *unbalanced* short-circuit calculations are similar, but evaluated per phase.

Note: Due to the way the equations were derived, they can only be used if the capacitor is lossless ($G_{cap} = 0$) and has no inherent coupling ($B0toB1 = 1$).

Since the linearised impedance is current dependant, the short-circuit needs to be solved iteratively. The iteration converges if the current change between two steps is smaller than the threshold set in the command (*e:curerr*).

To improve the convergence behaviour, the following was implemented:

- To avoid toggling between *conducting* and *non-conducting* states for currents close to the threshold a small hysteresis has been added. The MOV will switch to *conducting* if $I \geq 0{,}98 \cdot I_{pr}$, but will become *non-conducting* only if $I < 0{,}97 \cdot I_{pr}$.
- As the equations are prone to over- and under-shooting the target value during the iteration, a relaxation factor has been added to the short-circuit command (*e:iterRelax*) to speed up the convergence. The relaxation factor allows to control the portion of the current change that is used to calculate the linearised impedance for the next iteration step.

$$i_{pr} = \frac{\left|\underline{I}_{N-1}\right| + k_{relax} \cdot (|\underline{I}_N| - \left|\underline{I}_{N-1}\right|)}{I_{pr}}$$

3.1.3 ANSI Short-Circuit

Note: Series capacitors configured as single-phase capacitor models are not considered in any *PowerFactory* ANSI short-circuit calculation.

If the short-circuit command setting *Bypass Series Capacitor* is configured to a value other than "No Bypassing", series capacitors will be neglected (i.e. bypassed) for the selected calculations.

Depending on the calculation method used for the X/R calculation, the following applies:

Separate R and X Series capacitors are always neglected (i.e. bypassed);

Complex Series capacitors are considered according to the selection made for *Bypass Series Capacitor*.

3.1.4 IEC 61363

Series capacitors configured as 3-phase AC capacitor models are neglected (i.e. bypassed) in the IEC 61363 short-circuit calculation.

Note: Series capacitors configured as single-phase models are not considered in any *PowerFactory* IEC 61363 short-circuit calculation.

3.2 DC Model

Series capacitors configured as DC capacitor models are not considered in any *PowerFactory* short-circuit calculation.

4 RMS-Simulation

4.1 AC Model

The RMS-simulation and the Load Flow calculation both use the same circuit model representation of the series capacitance. Please refer to Section 2.1.

4.2 DC Model

In DC circuits, network transients are taken into consideration. The series capacitance is therefore modelled by the differential equation of a capacitance and a conductance in parallel:

$$i(t) = G \cdot u_{12}(t) + C \cdot \frac{du_{12}(t)}{dt} \tag{7}$$

where:

$$u_{12}(t) = u_1(t) - u_2(t)$$

5 EMT-Simulation

5.1 AC Model

5.1.1 Without Metal Oxide Varistor Protection

For EMT simulation, without any Metal Oxide Varistor (MOV) protection, the series capacitance is modelled by the differential equation of a capacitance and a conductance in parallel:

$$i(t) = G \cdot u_{12}(t) + C \cdot \frac{du_{12}(t)}{dt} \tag{8}$$

where:

$$u_{12}(t) = u_1(t) - u_2(t)$$

For a 3 phase balanced model without option *Variable capacitance and conductance via input signals* (see EMT page):

$$\begin{bmatrix} i_a(t) \\ i_b(t) \\ i_c(t) \end{bmatrix} = \begin{bmatrix} G_s & G_m & G_m \\ G_m & G_s & G_m \\ G_m & G_m & G_s \end{bmatrix} \cdot \begin{bmatrix} u_{12,a}(t) \\ u_{12,b}(t) \\ u_{12,c}(t) \end{bmatrix} + \begin{bmatrix} C_s & C_m & C_m \\ C_m & C_s & C_m \\ C_m & C_m & C_s \end{bmatrix} \cdot \begin{bmatrix} du_{12,a}(t)/dt \\ du_{12,b}(t)/dt \\ du_{12,c}(t)/dt \end{bmatrix} \tag{9}$$

where:

$$u_{12,a}(t) = u_{1,a}(t) - u_{2,a}(t)$$
$$u_{12,b}(t) = u_{1,b}(t) - u_{2,b}(t)$$
$$u_{12,c}(t) = u_{1,c}(t) - u_{2,c}(t)$$

and

$$G_s = (G_{0,cap} + 2 \cdot G_{cap})/3$$
$$G_m = (G_{0,cap} - G_{cap})/3$$

and:

$$C_s = (C_{0,cap} + 2 \cdot C_{cap})/3$$
$$C_m = (C_{0,cap} - C_{cap})/3$$

where $C_{0,cap} = B0toB1 \cdot C_{cap}$

When the model is unbalanced, no coupling between the phase are supported:

$$\begin{bmatrix} i_a(t) \\ i_b(t) \\ i_c(t) \end{bmatrix} = \begin{bmatrix} G_{cap,a} \\ G_{cap,b} \\ G_{cap,c} \end{bmatrix} \cdot \begin{bmatrix} u_{12,a}(t) \\ u_{12,b}(t) \\ u_{12,c}(t) \end{bmatrix} + \begin{bmatrix} C_{cap,a} \\ C_{cap,b} \\ C_{cap,c} \end{bmatrix} \cdot \begin{bmatrix} du_{12,a}(t)/dt \\ du_{12,b}(t)/dt \\ du_{12,c}(t)/dt \end{bmatrix} \tag{10}$$

The phase wise capacitance and conductance can be controlled via the corresponding input signals when the option *Variable capacitance and conductance via input signals* is enabled (see appendix A.2 and A.4). In addition the derivative of the capacitance input signals must be considered:

$$\begin{bmatrix} i_a(t) \\ i_b(t) \\ i_c(t) \end{bmatrix} = \begin{bmatrix} G_{cap,a} + dC_{cap,a}/dt \\ G_{cap,b} + dC_{cap,b}/dt \\ G_{cap,c} + dC_{cap,c}/dt \end{bmatrix} \cdot \begin{bmatrix} u_{12,a}(t) \\ u_{12,b}(t) \\ u_{12,c}(t) \end{bmatrix} + \begin{bmatrix} C_{cap,a} \\ C_{cap,b} \\ C_{cap,c} \end{bmatrix} \cdot \begin{bmatrix} du_{12,a}(t)/dt \\ du_{12,b}(t)/dt \\ du_{12,c}(t)/dt \end{bmatrix} \tag{11}$$

Note: In case when the $B0toB1$ or the $G0toG1$ ratio is different when 1 and the option *Variable capacitance and conductance via input signals* is enabled a warning message is issued *'The B0/B1 ratio and G0/G1 ratio are set to 1 for models with variable capacitance and conductance (no coupling between the phases).'*. Also if the capacitance input signals are not constant at the beginning of the simulation, this can lead to a transient, i.e. the model cannot be initialised exactly.

5.1.2 With Metal Oxide Varistor Protection

When the option *Metal Oxide Varistor* is selected, a non linear resistance can be defined which represents the protection of series capacitances by the MOV. The nonlinear resistance is entered by a voltage-current characteristic table. The entire characteristic is then approximated using cubic splines or alternative with a piecewise linear interpolatio. The corresponding equivalent circuit diagram is shown in Figure 5.1.

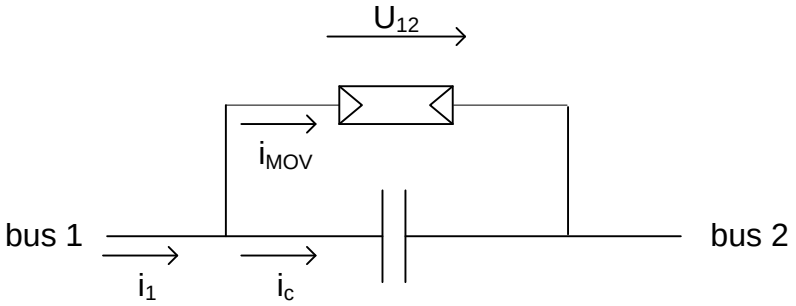


Figure 5.1: Series capacitance with MOV protection. The parallel conductance, not shown in the figure, is also included in the model

The model equations of the series capacitance with MOV protection are:

$$i_{MOV}(t) = f(u_{12}(t)) \tag{12}$$

$$i_c(t) = G \cdot u_{12} + C \cdot \frac{du_{12}}{dt} \tag{13}$$

$$i_1 = i_{MOV} + i_c \tag{14}$$

The dissipated MOV energy is calculated as follows:

$$P_{MOV}(t) = i_{MOV}(t) \cdot u_{MOV}(t) \tag{15}$$

$$E_{MOV}(t) = \int_{tstart}^{t} P_{MOV}(\tau) d\tau \tag{16}$$

5.1.3 Spark Gap

The series capacitor model for EMT simulations allows the definition of a protection spark gap, via the selection of the *Spark Gap* option. Once this option is selected, the resistance and reactance of the damping circuit can be defined. The spark gap model with optional parallel MOV surge arrester is shown in Figure 5.2.

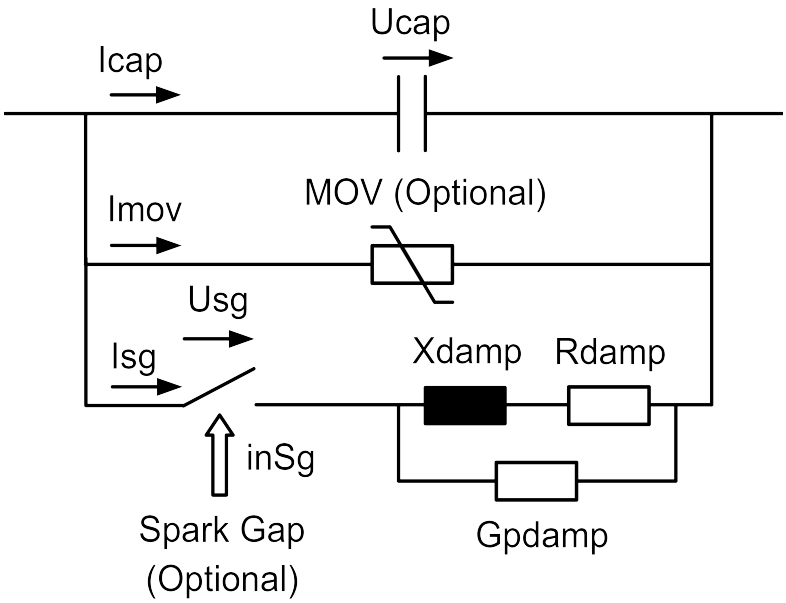


Figure 5.2: Spark gap model. The parallel conductance, not shown in the figure, is also included in the model

External Tripping

When the option *Built-In Tripping Unit* is deactivated, the spark gap can be triggered via the use of input signals *inSg_A*, *inSg_B* and *inSg_C* (defined in Table A.1) using either a parameter event (*EvtParam*) defined for the EMT simulation, or via an external DSL controller. The surge arrester is described in detail in the *PowerFactory* Surge Arrester Technical Reference document.

Built-In Tripping Unit

By activating the option *Built-In Tripping Unit*, it is possible to specify when to trigger the spark gap, according to any of the following thresholds:

- Trip when Current value is above the specified value in kA
- Trip when Voltage value is above the specified value in kV. Available only for option *Metal Oxide Varistor* deactivated.
- Trip when Energy value is above the specified value in MW. Available only for option *Metal Oxide Varistor* selected.
- Trip when Power value is above the specified value in MW. Available only for option *Metal Oxide Varistor* selected.

Concurrent Tripping

The triggering of the models can be done concurrently by the built-in tripping conditions as well as by using the external signals.

5.2 DC Model

In DC systems, the EMT model is identical to the RMS model. The DC model does not consider non linear resistance.

6 Harmonics/Power Quality

6.1 Frequency-Dependent Characteristics

If the *Balanced* input option is used, frequency-dependent characteristics may be defined for the capacitance C ($ccap$), conductance G and zero sequence capacitance and conductance. If the *Unbalanced* input option is used, frequency-dependent characteristics may be defined for the capacitance C per-phase ($ccap_a$, $ccap_b$ and $ccap_c$) and conductance per-phase.

Note: For absolute characteristics, the values defined in the element (not in the characteristic) will be used at the fundamental frequency. In addition, for an *Unbalanced* capacitor, only *relative* characteristics may be defined (i.e. not characteristics which are set to use *absolute* values).

The B_{cap} and G_{cap} used in the harmonics studies are calculated depending on the defined characteristic in case of *Balanced* input option as:

$$
\begin{aligned}
B_{cap}(\omega_n \cdot h) &= f(ccap, \omega_n \cdot h) \\
G_{cap}(\omega_n \cdot h) &= f(bcap \cdot tandc, \omega_n \cdot h) \\
B_{0,cap}(\omega_n \cdot h) &= f(B_{cap}(\omega_n) \cdot B0toB1, \omega_n \cdot h) \\
G_{0,cap}(\omega_n \cdot h) &= f(G_{cap}(\omega_n) \cdot G0toG1, \omega_n \cdot h)
\end{aligned}
$$

and in case of *Unbalanced* input option as:

$$
\begin{aligned}
B_{a,cap}(\omega_n \cdot h) &= f(ccap_a, \omega_n \cdot h) \\
G_{a,cap}(\omega_n \cdot h) &= f(bcap_a \cdot tandc, \omega_n \cdot h) \\
B_{b,cap}(\omega_n \cdot h) &= f(ccap_b, \omega_n \cdot h) \\
G_{b,cap}(\omega_n \cdot h) &= f(bcap_b \cdot tandc, \omega_n \cdot h) \\
B_{c,cap}(\omega_n \cdot h) &= f(ccap_c, \omega_n \cdot h) \\
G_{c,cap}(\omega_n \cdot h) &= f(bcap_c \cdot tandc, \omega_n \cdot h)
\end{aligned}
$$

where ω_n is the nominal fundamental frequency of the corresponding grid and h is the harmonic order.

In case of *Balanced* input option, the equations are:

$$
\begin{aligned}
\underline{I}_{1,bus1} &= (G_{cap}(\omega_n \cdot h) + jB_{cap}(\omega_n \cdot h)) \cdot \underline{U}_{1,12} \\
I_{1,bus1} + I_{1,bus2} &= 0 \\
\underline{I}_{2,bus1} &= (G_{cap}(\omega_n \cdot h) + jB_{cap}(\omega_n \cdot h)) \cdot \underline{U}_{2,12} \\
I_{2,bus1} + I_{2,bus2} &= 0 \\
\underline{I}_{0,bus1} &= (G_{0,cap}(\omega_n \cdot h) + jB_{0,cap}(\omega_n \cdot h)) \cdot \underline{U}_{0,12} \\
I_{0,bus1} + I_{0,bus2} &= 0
\end{aligned}
$$

In case of *Unbalanced* input option the equations are:

$$
\begin{aligned}
\underline{I}_{a,bus1} &= (G_{a,cap}(\omega_n \cdot h) + jB_{a,cap}(\omega_n \cdot h)) \cdot \underline{U}_{a,12} \\
\underline{I}_{b,bus1} &= (G_{b,cap}(\omega_n \cdot h) + jB_{b,cap}(\omega_n \cdot h)) \cdot \underline{U}_{b,12} \\
\underline{I}_{c,bus1} &= (G_{c,cap}(\omega_n \cdot h) + jB_{c,cap}(\omega_n \cdot h)) \cdot \underline{U}_{c,12} \\
\underline{I}_{a,bus1} + \underline{I}_{a,bus2} &= 0 \\
\underline{I}_{b,bus1} + \underline{I}_{b,bus2} &= 0 \\
\underline{I}_{c,bus1} + \underline{I}_{c,bus2} &= 0
\end{aligned}
$$

A Signals Definitions

Variable	Unit	I/O	Description
Emov_a	MWs	STATE	MOV absorbed energy, A
Emov_b	MWs	STATE	MOV absorbed energy, B
Emov_c	MWs	STATE	MOV absorbed energy, C
inSg_A		IN	Spark Gap, A
inSg_B		IN	Spark Gap, B
inSg_C		IN	Spark Gap, C

Table A.1: Internal signals (EMT-Simulation) for a 3 phase model

When option *Variable capacitance and conductance via input signals* is enabled the following additional input signals are available:

Variable	Unit	I/O	Description
Cin_A	F	IN	Capacitance (Input), A
Cin_B	F	IN	Capacitance (Input), B
Cin_C	F	IN	Capacitance (Input), C
Gin_A	S	IN	Conductance (Input), A
Gin_B	S	IN	Conductance (Input), B
Gin_C	S	IN	Conductance (Input), C

Table A.2: Additional signals (EMT-Simulation) for a 3 phase model

And for a single phase or DC model:

Variable	Unit	I/O	Description
Emov	MWs	STATE	MOV absorbed energy
inSg (only for AC model)		IN	Spark Gap

Table A.3: Internal signals (EMT-Simulation) for a 1 phase or DC model

When option *Variable capacitance and conductance via input signals* is enabled the following additional input signals are available:

Variable	Unit	I/O	Description
Cin	F	IN	Capacitance (Input)
Gin	S	IN	Conductance (Input)

Table A.4: Additional signals (EMT-Simulation) for a 1 phase or DC model

B Parameter Definitions

Parameter	Unit	Default Value	Description	Range	Symbol
loc_name			Name		
bus1			Terminal i (StaCubic)		
bus1_bar			Terminal i		
bus2			Terminal j (StaCubic)		
bus2_bar			Terminal j		
iZoneBus		0	Zone		
cpZone			Zone		
iAreaBus		0	Area		
cpArea			Area		
outserv		0	Out of Service	x≥0 and x≤1	
ucn	kV	1,	Ratings: Rated Voltage	x≥0	
Sn	MVA	1,732051	Ratings: Rated Power	x>0	
Curn	kA	1,	Ratings: Rated Current	x>0	
pRating			Thermal Rating (IntThrating)		
systp		AC	System Type	AC:DC	
nphases		3	Phases	1:3	
Inom	kA	1,	Rated Current		
i_sym		0	Balanced/Unbalanced	x=0 or x=1	
ccap	F	0,	Value: Capacitance, C	x≥0	C
bcap	S	0,	Value: Susceptance, B	x≥0	
xcap	Ω	0,	Value: Reactance 1/B	x≥0	
tandc		0,	Value: Loss Factor, tan(delta)	x≥0	
gcap	S	0,	Value: Conductance, G	x≥0	
bcap_a	S	0,	Value / Phase A: Susceptance, B	x≥0	
ccap_a	F	0,	Value / Phase A: Capacitance, C	x≥0	
xcap_a	Ω	#INF	Value / Phase A: Reactance 1/B	x≥0	
bcap_b	S	0,	Value / Phase B: Susceptance, B	x≥0	
ccap_b	F	0,	Value / Phase B: Capacitance, C	x≥0	
xcap_b	Ω	#INF	Value / Phase B: Reactance 1/B	x≥0	
bcap_c	S	0,	Value / Phase C: Susceptance, B	x≥0	
ccap_c	F	0,	Value / Phase C: Capacitance, C	x≥0	
xcap_c	Ω	#INF	Value / Phase C: Reactance 1/B	x≥0	

B0toB1		1,	Value: B0/B1 ratio	x≥0	
G0toG1		1,	Value: G0/G1 ratio	x≥0	
maxload	%	100,	Thermal Loading Limit: Max. Loading	x≥0	
iSparkGap		0	Spark Gap	x=0 or x=1	
Rdamp	Ω	0,	Spark Gap Damping Circuit: Resistance	x≥0	Rdamp
Xdamp	Ω	0,	Spark Gap Damping Circuit: Reactance	x≥0	Xdamp
Gpdamp	S	0,	Spark Gap Damping Circuit: Parallel- Conductance	x≥0	Gpdamp
i_enter		0	Metal Oxide Varistor	x=0 or x=1	
Im	kA	0,	Current		I_{MOV}
Vm	kV	0,	Voltage		

Table B.1: Input parameters for series capacitor element

Parameter	Unit	Default Value	Description	Range	Symbol
iInterPol		0	Interpolation	spline:-piecewise linear	
smoothfac	%	10,	Smoothing Factor	x≥0.0 and x≤100.0	
gnrl_modif		01.01.1970 01:00:00	Object modified		
gnrl_modby			Object modified by		
sernum			Serial Number		
manuf			Manufacturer		
constr		0	Year of Construction		
iComDate		01.01.1970 01:00:00	Commissioning Date		
chr_name			Characteristic Name		
for_name			Foreign Key		
dat_src		MAN	Data source		
doc_id			Additional Data ()		
pOwner			Owner (ElmOwner)		
pOperator			Operator (ElmOperator)		
desc			Description		
appr_status		Not Approved	Approval Information: Status		
appr_modif		01.01.1970 01:00:00	Approval Information: Modified		
appr_modby			Approval Information: Modified by		
cimRdfId			RDF ID		

Table B.2: Input parameters for series capacitor element

C References

[1] Daniel L. Goldsworthy. A linearized mode for mov-protected series capacitors. *IEEE Transactions on Power Systems Vol PWRS-2, No. 4*, 1987.

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